

RAHUL MITRA

CYBER SECURITY PROFILE

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PROFILE SUMMARY

Early-career cyber security engineer and bug bounty researcher with hands-on vulnerability discovery across Singapore government and transport programs. Combines adversarial testing, automation scripting, secure design awareness, and software engineering depth.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Software Engineering; Artificial Intelligence; 3D Computer Vision; Computer Systems; Statistics; Mathematics; Engineering Systems
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERSHIP EXPERIENCE

Independent Bug Bounty Researcher, YesWeHack Platform

May 2024 - Present

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Used Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts to automate endpoint discovery, payload delivery, and evidence collection for responsible disclosure reports.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Jun 2026

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

SELECTED PROJECTS AND COMPETITIONS

AsyncDDGS - Open-Source Search Client

2024 - Present

- Maintained an asyncio-first, aiohttp-based DuckDuckGo search client on PyPI with pytest CI/CD and adoption across AI tooling, Discord bots, and search-enabled projects.

Arcane Security Tooling Family

2026

- Built Rust-forward security and red-team tooling experiments spanning Arcane, Arcane-PP, Arcane-GLM, and Arcane-OCR for automation, OCR, and adversarial workflows.

DSTA BRINHACK 2025 CTF

2025

- Placed in the top quartile among 400+ teams in a first CTF, applying web, crypto, reverse engineering, and systems security problem-solving under time pressure.

AgeWellLah.AI - NUS HealthHack 2025

2025

- Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Burp Suite; Wireshark; SSRF; CSRF; SQL Injection; NoSQL Injection; Authentication Bypass; OAuth2/JWT; OWASP Top 10; Python; Bash; Rust; Fuzzing; Reverse Engineering.

- Reverse Engineering Windows Executables - digital forensics and executable analysis.
- Docker for the Absolute Beginner - container fundamentals and deployment workflows.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.